

Abstract

This graduation project aims to optimize the management of student data, enhance accessibility, and alleviate the administrative burden on staff members. It involves re-implementing the previous year's graduation project, 'FCAI website (student's section),' with a focus on enhancing its performance and effectiveness. Through the utilization of innovative solutions, we aspire to contribute to the enhancement of the institution's operational efficiency and overall user experience by refining design and functionality to better align with the needs of the student affairs department.

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Chapter 1 Introduction

The primary objective of this project is to develop an internal system for the student affairs department that reduces the staff's workload and automates certain tasks, thereby simplifying, and expediting their job responsibilities in an efficient manner. Additionally, we plan to integrate our system with another existing system (new E-com graduation project developed in 2023). However, after careful assessment, which included testing the existing system, we have found that integrating with it would require a significant amount of time and may not be feasible within a given deadline. Therefore, we have decided to develop this system from scratch to address its deficiencies effectively.

1.1 Background and Motivation

The student affairs staff faces a heavy manual workload in managing students and departmental operations, posing challenges in terms of time, resources, and potential errors. These tasks need to be automated to ease their workload, minimize errors, and enhance departmental efficiency.

1.2 Problem Statement

Current manual processes within the student affairs department create barriers to efficiency and effectiveness. These manual processes hinder productivity and increase the likelihood of errors. The department's ability to provide optimal support and services to students is compromised.

1.3 Objectives

- Develop and integrate automation features within the student affairs internal system to reduce manual workload.
- Enhance system security and compliance with data protection regulations.

1.4 Project Scope and Limitations

Scope:

- Automation Features: The project will focus on developing and integrating automation features into the student affairs internal system. These features will target specific tasks and processes that are currently performed manually, with the aim of reducing the workload of the staff.
- Enhanced Security: Alongside automation, the project will also enhance the security of the system. This includes implementing measures to protect sensitive student data and ensuring compliance with relevant data protection regulations.
- Improved Analytical Reporting: The project will streamline and optimize the process of generating analytical reports. Currently, creating reports is a laborious and time-consuming task. The project will introduce features and tools to make the generation of reports more efficient, enabling faster and more effective data analysis for better decision-making.

Limitations:

- Resource Limitations: The project will be subject to resource constraints, including budget, time, and available technology. The extent of automation and security enhancements will depend on the resources allocated.
- Integration Challenges: Integration of automation features may face challenges related to compatibility with existing systems and software. These challenges could impact the timeline and effectiveness of the project.
- Training and Adoption: The success of the project relies on staff adoption of the automated system. However, training and change management may pose limitations in terms of staff acceptance and adaptation to the new system.
- Data Migration: Transferring existing data from manual records to the automated system may be a complex task. Data migration challenges could affect the accuracy and completeness of the digital records.
- Scope Creep: To maintain project focus and efficiency, it's important to define clear boundaries for the scope. Scope creep, which involves the unplanned expansion of project goals, will be managed and minimized.
- User Feedback and Iteration: While the project aims to address current needs,

evolving user requirements may arise during its development. The ability to accommodate changes and feedback within the defined scope will be considered.

- Testing and Quality Assurance: Thorough testing is crucial to ensure that the automated system operates reliably and accurately. Limitations in testing resources or unforeseen technical issues could impact the project's timeline.

The project will strive to address these limitations while staying within the defined scope to achieve its objectives of reducing manual workload, enhancing system security, and improving efficiency within the student affairs department.

1.5 Project Plan (Gantt Chart)

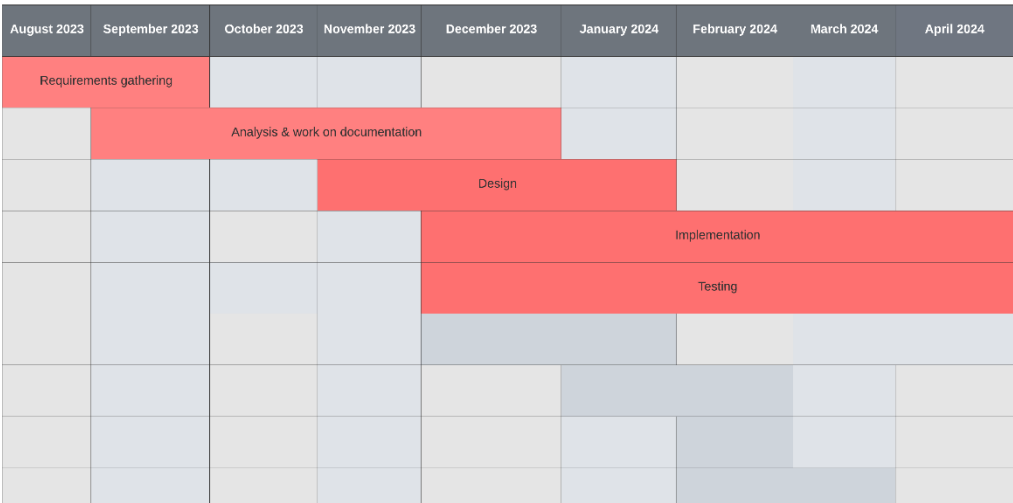


Figure 1. Gantt Chart

1.6 Project Methodology: Agile

We will choose the Agile methodology as the preferred approach to manage the project aimed at improving the student affairs internal system. This choice is driven by the following reasons:

- **Flexibility and Adaptability:** The nature of this project involves continuous improvement and evolving requirements.
- **Stakeholder Collaboration:** Encourages close collaboration with stakeholders, including student affairs staff and system users. This collaboration ensures that their input is actively considered throughout the project, resulting in a system that better

aligns with their needs.

- **User-Centric Focus:** Place a strong emphasis on delivering value to end-users promptly. This aligns perfectly with the goal of improving the student affairs internal system to enhance user experience and efficiency.
- **Early and Frequent Deliveries:** Promotes the delivery of small, functional increments at regular intervals. This approach allows us to provide tangible results quickly, keeping stakeholders engaged and informed about project progress.

By choosing agile as our project methodology, we are not only embracing its inherent flexibility and stakeholder collaboration but also leveraging its capacity for internal iterative cycles. This approach will enable us to adapt to changing requirements, prioritize user needs, and refine our deliverables.

1.7 Project Report Outline

This document outline will be as follows:

- Chapter 2 presents the literature review of the project.
- Chapter 3 presents the analysis of the Student Affairs.
- Chapter 4 presents an explanation of the implementation and design of the project.
- Chapter 5 presents test cases and project evaluation.
- Chapter 6 concludes the work and lessons learned, then sheds light to future work.

Chapter 2 Market and Literature Survey

2.1 SWOT analysis

Strengths:

- Improved Efficiency: The system streamlines manual processes, reducing the workload and increasing the efficiency of student affairs staff.
- User-Focused: Enhancing the experience for both staff and students.
- Flexibility: Agile methodology allows for adaptability to changing requirements, ensuring the system remains up to date.
- Collaboration: Stakeholder collaboration is encouraged, leading to a system that aligns well with the needs of the department.

Weaknesses:

- Implementation Challenges: Transitioning to an automated system may pose challenges, including staff training and potential resistance to change.
- Resource Intensive: Developing and implementing the system may require significant resources, both in terms of time and finance.
- Data Migration: Migrating existing data to the new system may be complex and time-consuming, potentially causing data integrity issues.
- Integration Complexity: Integrating the system with other existing systems or databases may present technical challenges.

Opportunities:

- Continuous Improvement: Agile methodology allows for ongoing refinement and adaptation, ensuring the system evolves to meet changing needs.
- Data-Driven Decision-Making: The system's enhanced analytics capabilities offer opportunities to make data-driven decisions and improve services.
- Increased User Adoption: As the system becomes more user-friendly and efficient, it can lead to higher user adoption rates.
- Expanded Services: The system can be expanded to include new services or features to meet emerging needs.

Threads:

- Technical Challenges: Technical issues or system failures could disrupt operations and negatively impact user satisfaction.
- Data Security: The system must address potential security vulnerabilities to protect sensitive student data.

- User Resistance: Resistance to change from staff or students could impede the successful implementation and adoption of the system

Chapter 3 Student Affairs Analysis

3.1 Functional Requirements

- Automate student registration.
- Integrate CKES with new E-com.
- Reporting and analytics

3.2 Non-Functional Requirements

Performance Requirements:

- The system should be able to handle concurrent user sessions without significant degradation in response times, with a response time of less than 2 seconds for common operations.

Usability Requirements:

- The system should have an intuitive and user-friendly interface to ensure that both technical and non-technical staff can easily navigate and use the system without extensive training.

Security Requirements:

- The system must implement strong user authentication and authorization mechanisms to protect sensitive student data.

Availability Requirements:

- The system should be available 24/7, with a maximum allowable downtime of no more than 1 hour per month, scheduled during off-peak hours.

Adaptability Requirements:

- The system should be able to accommodate changes in student affairs processes and policies without requiring extensive code changes and support configuration changes through an admin interface.

Reliability Requirements:

- It should have automated backups and disaster recovery procedures in place.

Compatibility Requirements:

- The system should be compatible with commonly used databases and operating systems.

3.3 Stakeholders:

- Student affairs Department/Administrators: Oversee the website's development, maintenance, and content management. Ensure that the website aligns with the department's goals and objectives. Approve and update content as needed.
- Vice Dean: Approval of Student Requests / Accessing Reports and Data / Ensuring that approval decisions align with institutional policies, academic standards, and departmental guidelines.
- Center of knowledge & Electronic Services (CKES)
- Students

3.4 Use Case Diagram

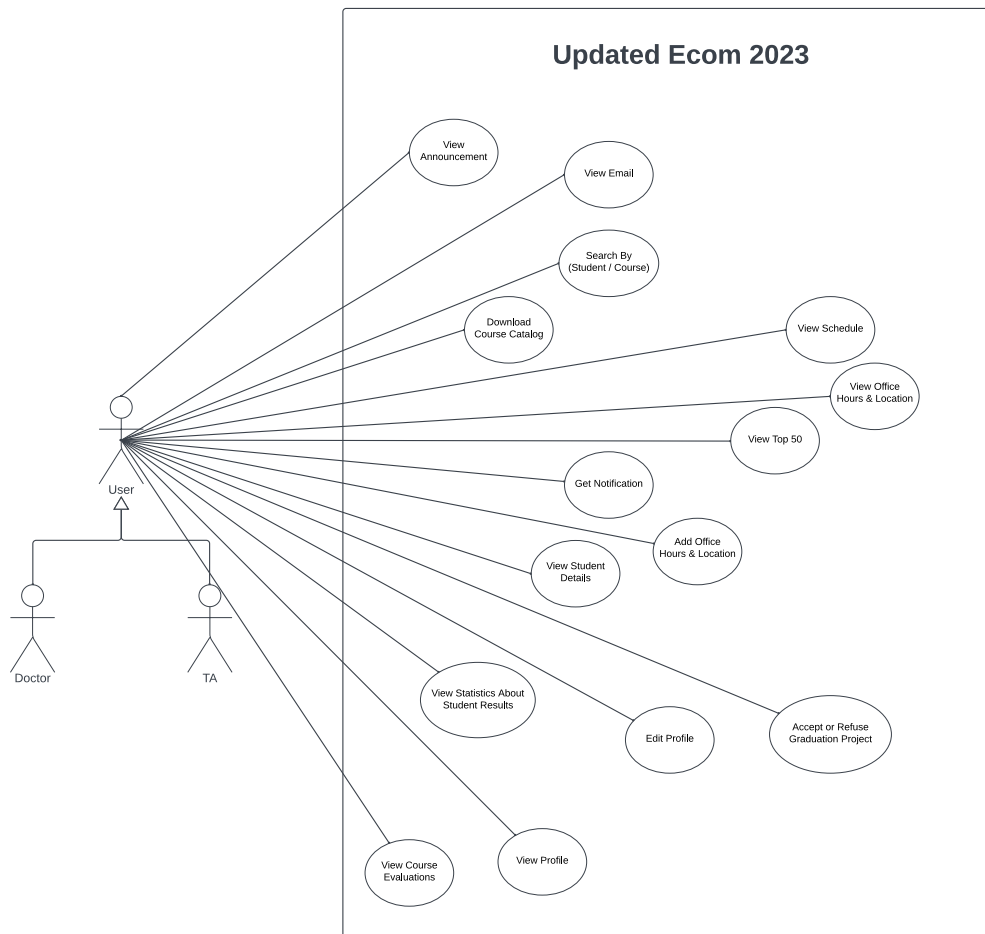


Figure 2. Use Case Diagram for Professor and TA (Ecom 2023)

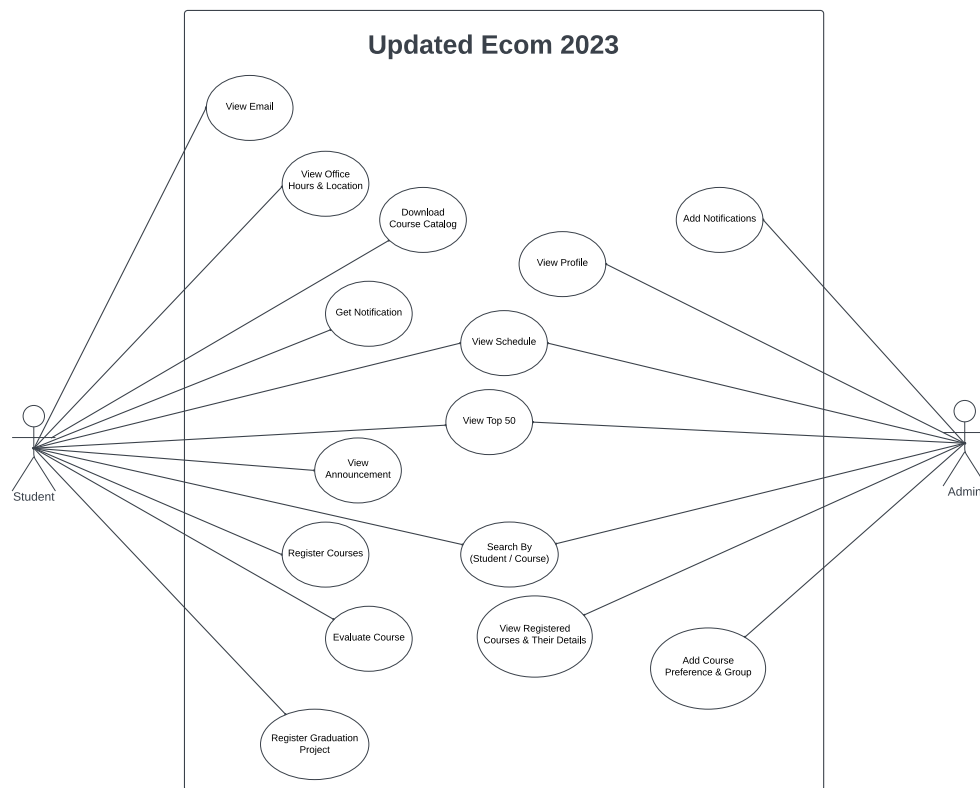


Figure 3. Use Case Diagram for Student and Admin (Ecom 2023)

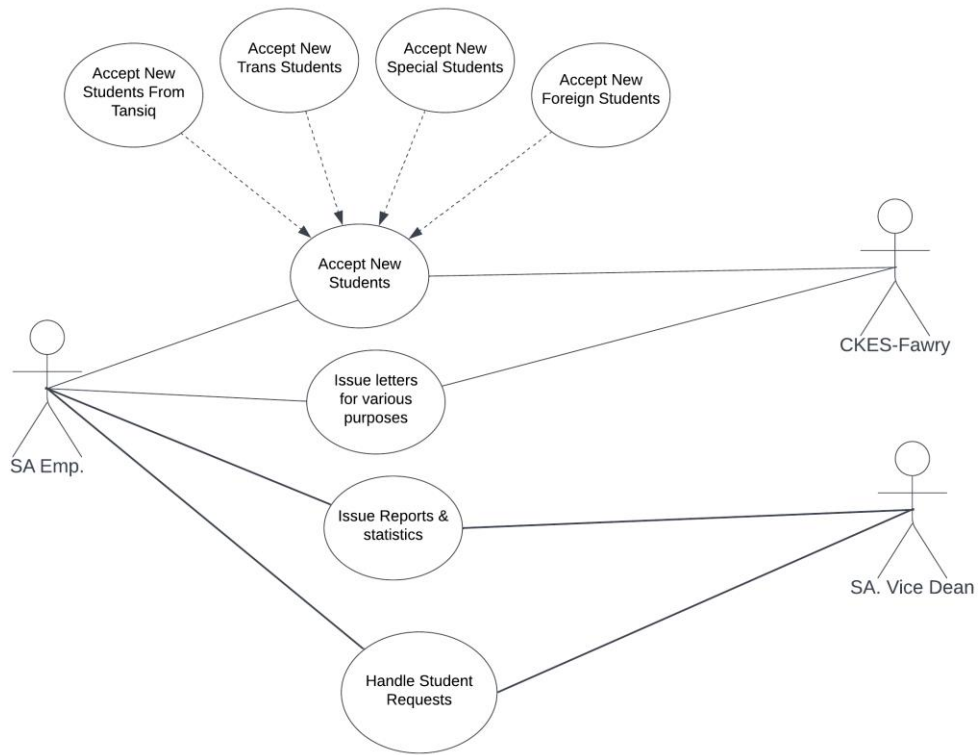


Figure 4. Student Affairs Use Case Diagram

3.5 Use Case Description

3.5.1 Register Course (Ecom 2023)

USE CASE NAME	Register Course	
SCENARIO	When the student receives a notification that the registration is opened the student enters to select his courses and their groups.	
TRIGGERING EVENT	When the registration opens.	
BRIEF DESCRIPTION	When the student receives a notification that the registration is opened the student enters and shows a list of available courses and then selects his courses and their groups.	
ACTORS	Student	
RELATED USE CASES	Login User	
STAKEHOLDERS	Student, Student Affairs, Professor	
PRECONDITIONS	Students must have an existing account.	
POSTCONDITIONS	Both the related courses and their groups are updated.	
FLOW OF ACTIVITIES	Actor	System
	1. The student opens his/her registration page.	1.1. System displays all available courses for this student.
	2. The student selects his/her courses.	2.1. System checks the limit of credit hours of this user.
	3. The student selects his/her groups.	3.1. System assigns to a group. 3.2. System returns the schedule.
EXCEPTION CONDITIONS	2.1 Students reach the minimum or maximum credit hours' limit. 3.1 There are conflict between the groups or courses.	

3.5.2 Push Notification (Ecom 2023)

USE CASE NAME	Push Notification	
SCENARIO	When the Professor or Student Affairs opens the registration, publishes results of exams, schedule is released or there is an announcement.	
TRIGGERING EVENT	When there is new news about results, schedule, or registration.	
BRIEF DESCRIPTION	When the registration opens, there is an announcement or the schedule is released, A notification will be sent to students and also will be sent to the professor and TA if the results of feedback of the students are ready or there is an announcement, or the schedule is released.	
ACTORS	Student Affairs	
RELATED USE CASES	Login User	
STAKEHOLDERS	Professor, Student Affairs.	
PRECONDITIONS	Student Affairs must upload the schedule, results or open the registration.	
POSTCONDITIONS	Students, Professors, or TAs go to the site and follow the notification.	
	Actor	System
FLOW OF ACTIVITIES	1. The Student Affairs upload the schedule, results or open the registration. 1.1. System push notification to TA, Professor, and student. 2. The TA, Professor, and student received the notification.	
EXCEPTION CONDITIONS	1.1. Student, professor, or TA doesn't activate the notification feature.	

3.5.3 View Profile (Ecom 2023)

USE CASE NAME	View Profile	
SCENARIO	User could see his/her profile.	
TRIGGERING EVENT	When the user wants to check his/her information and edit it if he/she wants.	
BRIEF DESCRIPTION	When the registration opens or the schedule is released, there is an announcement. A notification will be sent to students as well as the professor and TA if the results of feedback of the students are ready or there is an announcement, or the schedule is released.	
ACTORS	Student, professor, and TA	
RELATED USE CASES	Login User	
STAKEHOLDERS	Student, professor, TA, and student affairs.	
PRECONDITIONS	User must have an existing account	
POSTCONDITIONS	---	
FLOW OF ACTIVITIES	Actor	system
	1. User open his /her profile	1.1 System display User information
EXCEPTION CONDITIONS	---	

3.5.4 Register Graduation Project (Ecom 2023)

USE CASE NAME	Register Graduation Project	
SCENARIO	When the registration of the graduation project opens, the student can register the graduation project and put the idea of it, then select from available professors and TAs from his/her department.	
TRIGGERING EVENT	When the registration of the graduation project opens.	
BRIEF DESCRIPTION	When the student receives a notification that the registration is opened the student enters the project idea then selects from available professors and TAs from his/her department.	
ACTORS	Student, Professor, and TA	
RELATED USE CASES	Login User and Get Notification.	
STAKEHOLDERS	Student, professor, TA, and student affairs.	
PRECONDITIONS	Students must have an existing account.	
POSTCONDITIONS	The student will get a reply from the professor or TA.	
FLOW OF ACTIVITIES	Actor	System
	1. The student sends a description about the graduation project to the professor. 2. Professor accepts that team. 3. The student could view the acceptance.	1.1. System sends it to the Professor. 2.1. System sends the acceptance to the team.
EXCEPTION CONDITIONS	2. Professor refuses that team.	

3.5.5 Evaluate Course (Student's Side) (Ecom 2023)

USE CASE NAME	Evaluate Course (Student's Side)	
SCENARIO	When the student receives an announcement about the course feedback that he/she can choose each course and puts his/her feedback.	
TRIGGERING EVENT	When the Feedback of content of courses, professors and TAs is opened.	
BRIEF DESCRIPTION	When the student receives an announcement about the course feedback that he/she can choose each course and put his/her feedback about the professor, content, and the TA.	
ACTORS	Students	
RELATED USE CASES	Login User, Get Notification (for making the feedback).	
STAKEHOLDERS	Student, professor, and TA.	
PRECONDITIONS	Students must have an existing account and register for that course.	
POSTCONDITIONS	Feedback sent to Professors, TAs.	
FLOW OF ACTIVITIES	Actor	System
	1. The student receives notification that he/she could write his/her feedback about courses. 2. Students enter to give his/her feedback about the content of the courses, professors, and TAs. 3. Professor and TA receives the feedback.	2.1. System sends the feedback to the professor and TA without the student's information.
EXCEPTION CONDITIONS	2.2 Student is not registered in this course.	

3.5.6 Chat (Ecom 2023)

USE CASE NAME	Chat	
SCENARIO	The student can chat with his/her professor and TA.	
TRIGGERING EVENT	When the student wants to connect with his/her professor or TA.	
BRIEF DESCRIPTION	The student can chat with his/her professor and TA to ask about the content he/she doesn't understand or any other question.	
ACTORS	Student, professor, and TA.	
RELATED USE CASES	Login User.	
STAKEHOLDERS	Student, professor, and TA.	
PRECONDITIONS	Students must have an existing account.	
POSTCONDITIONS	Students understand missing things for them.	
FLOW OF ACTIVITIES	Student	Professor or TA
	1. The student can start chatting with his/her professor or TA.	1.1 The professor or TA receives the message and answers it when he/she is available.
EXCEPTION CONDITIONS	2. Students are not logged in the system.	

3.5.7 View Office Hours and Locations (Ecom 2023)

USE CASE NAME	View Office Hours and Location	
SCENARIO	The student can view the office hours of professors and TAs can also view the location of their offices.	
TRIGGERING EVENT	---	
BRIEF DESCRIPTION	The student can view the office hours of professors and TAs can also view the location of their offices.	
ACTORS	Student, professor, and TA.	
RELATED USE CASES	Login User.	
STAKEHOLDERS	Student Affairs, professor, and TA.	
PRECONDITIONS	Students must have an existing account.	
POSTCONDITIONS	A report of office hours of professors and TAs with their location.	
FLOW OF ACTIVITIES	Actor	System
	1. The student opens the office hours page.	1.1. System displays all office hours of professors and TAs and their location.
EXCEPTION CONDITIONS	1.1 Can be no office hours assigned till a specific due date or not specified ever.	

3.5.8 View Announcements (Ecom 2023)

USE CASE NAME	View Announcements						
SCENARIO	When the faculty has new news to tell the students about it like the schedule uploaded, training is opened and so on.						
TRIGGERING EVENT	When the faculty makes that announcement.						
BRIEF DESCRIPTION	When the faculty has new news to tell the students about it like Schedule uploaded, training is opened and so on, when the announcement is published the students can see it.						
ACTORS	Student.						
RELATED USE CASES	Login User.						
STAKEHOLDERS	Student, professor, TA, and student affairs.						
PRECONDITIONS	Students must have an existing account.						
POSTCONDITIONS	The students can see the announcement.						
FLOW OF ACTIVITIES	<table border="1"> <thead> <tr> <th>Actor</th><th>System</th></tr> </thead> <tbody> <tr> <td>1. The student Affairs make the announcement.</td><td>1.1. System published the announcement on the site</td></tr> <tr> <td>2. The student receives the announcement and knows the news from it.</td><td></td></tr> </tbody> </table>	Actor	System	1. The student Affairs make the announcement.	1.1. System published the announcement on the site	2. The student receives the announcement and knows the news from it.	
Actor	System						
1. The student Affairs make the announcement.	1.1. System published the announcement on the site						
2. The student receives the announcement and knows the news from it.							
EXCEPTION CONDITIONS	---						

3.5.9 Report Statistics (About student Results) (Ecom 2023)

USE CASE NAME	Report Statistics (About Student Results)	
SCENARIO	When the professor/TA wants to see some statistics about the students results he/she selects the tap of statistics results.	
TRIGGERING EVENT	---	
BRIEF DESCRIPTION	When the professor/TA wants to see some statistics about the students' results he/she selects the tap of statistics results like (average students result, top course grades, etc..) and selects the course, level, or the department he/she wants then he/she can select the view he/she wants to see diagrams such (charts, histogram, ...).	
ACTORS	Professor and TA	
RELATED USE CASES	Login User, Register Course	
STAKEHOLDERS	Student affairs, professors, and TAs.	
PRECONDITIONS	Students must have an existing account.	
POSTCONDITIONS	A report of statistics about students' results.	
FLOW OF ACTIVITIES	Actor	System
	1. The professor/TA selects the tab of student results statistics. 2. The professor/TA can filter the report by course, level, or department.	1.1. System displays all course results. 2.1. System displays a report of statistics.
EXCEPTION CONDITIONS	---	

3.5.10 View Top 50 (Ecom 2023)

USE CASE NAME	View Top 50	
SCENARIO	view top 50 students filtered by level, department, or course.	
TRIGGERING EVENT	---	
BRIEF DESCRIPTION	When the student, professor or TA wants to know the Top 50 students in specific (level, department, or course) they could select the Top 50 page and filter by what they want.	
ACTORS	Student, professor, and TA.	
RELATED USE CASES	Login User	
STAKEHOLDERS	Student, professor, TA and student affairs.	
PRECONDITIONS	Students must have an existing account.	
POSTCONDITIONS	A report of students' IDs, names, and their GPAs.	
FLOW OF ACTIVITIES	Actor	System
	1. The user opens the top 50's page. 2. User selects to filter by (level, department, or course).	2.1. System sends a report of the student's IDs, names, and their GPAs.
EXCEPTION CONDITIONS	---	

3.5.11 View Student's Data (Ecom 2023)

USE CASE NAME	View Student's Data				
SCENARIO	View students filtered by level, department, or course.				
TRIGGERING EVENT	---				
BRIEF DESCRIPTION	When The professor, TA wants to know students in specific (names, GPA, grade in a specific course the professor is assigned to) they could select a page and filter by what he/she wants.				
ACTORS	Professor, TA, Student Affairs.				
RELATED USE CASES	Login User, Register Course.				
STAKEHOLDERS	Professor, TA, and Student Affairs.				
PRECONDITIONS	Students must have an existing account.				
POSTCONDITIONS	A report of Students IDs, names, GPAs, and grades in each course.				
FLOW OF ACTIVITIES	<table> <tr> <th>Actor</th><th>System</th></tr> <tr> <td>1. The professor, TA, wants to show information about specific student using search.</td><td>1.1. System shows the information about this student.</td></tr> </table>	Actor	System	1. The professor, TA, wants to show information about specific student using search.	1.1. System shows the information about this student.
Actor	System				
1. The professor, TA, wants to show information about specific student using search.	1.1. System shows the information about this student.				
EXCEPTION CONDITIONS	1.1 This Student has no account (not exist in the database of faculty).				

3.5.12 View a schedule of All Courses and All Levels to the professor and TAs (Ecom 2023)

USE CASE NAME	View a schedule of (all courses and all levels) to the Professor and TAs	
SCENARIO	When the professor or TA wants to view the schedule of all courses and all levels, he/she will enter the page of courses and select the level he/she wants to show.	
TRIGGERING EVENT	---	
BRIEF DESCRIPTION	The professor or TA wants to view the schedule of all courses and all levels, he/she will enter the page of courses and select the level he/she wants to show.	
ACTORS	Professor, TA.	
RELATED USE CASES	Login User	
STAKEHOLDERS	Professor, TA, and student affairs.	
PRECONDITIONS	The student affairs upload at least one schedule.	
POSTCONDITIONS	A report of a schedule of (all courses and all levels).	
FLOW OF ACTIVITIES	Actor	System
	1. The professor or TA selects the schedule page. 2. The professor or TA can choose to filter with level or course.	2.1 System returns the schedule of each course or each level depending on the filtration of the user.
EXCEPTION CONDITIONS	---	

3.5.13 View Course Evaluation (Ecom 2023)

USE CASE NAME	View Course Evaluation (Done by Students Anonymously)	
SCENARIO	When the professor or TA wants to see the feedback done by students without their names.	
TRIGGERING EVENT	When the Professor or TA receives an announcement that the students' feedback is available.	
BRIEF DESCRIPTION	When the professor, or TA wants to see the feedback done by students without their names after he/she receives an announcement that the students' feedback is available.	
ACTORS	Professor and TA.	
RELATED USE CASES	Login User and Send Feedback (Student's Side).	
STAKEHOLDERS	Professor and TA.	
PRECONDITIONS	Students must write their feedback.	
POSTCONDITIONS	A report of the students' feedback.	
FLOW OF ACTIVITIES	Actor	System
	1. The Professor or TA clicks on the announcement.	1.1. System opens a page with students' feedback without their names.
EXCEPTION CONDITIONS	---	

3.5.14 Download Schedule or Bylaw (Ecom 2023)

USE CASE NAME	Download Schedule or Bylaw				
SCENARIO	User wants to download the bylaw or schedule so he/she could know about courses the faculty offers and when the courses he/she want to register are held.				
TRIGGERING EVENT	Student Affairs add bylaw and schedule.				
BRIEF DESCRIPTION	Download bylaw or schedule.				
ACTORS	Student, professor, TA, and Student Affairs.				
RELATED USE CASES	Login User.				
STAKEHOLDERS	Student Affairs.				
PRECONDITIONS	---				
POSTCONDITIONS	---				
FLOW OF ACTIVITIES	<table> <tr> <th>Actor</th><th>System</th></tr> <tr> <td>1. User click in download button</td><td>1.1 System download the bylaw or schedule</td></tr> </table>	Actor	System	1. User click in download button	1.1 System download the bylaw or schedule
Actor	System				
1. User click in download button	1.1 System download the bylaw or schedule				
EXCEPTION CONDITIONS	---				

3.5.15 Search (Ecom 2023)

USE CASE NAME	Search	
SCENARIO	Users want to search by user or course.	
TRIGGERING EVENT	---	
BRIEF DESCRIPTION	Users want to search by user or course.	
ACTORS	Student, professor, TA, and Student Affairs.	
RELATED USE CASES	Login User and View profile	
STAKEHOLDERS	Student, professor, TA and Student Affairs.	
PRECONDITIONS	Users must have an existing account.	
POSTCONDITIONS	---	
FLOW OF ACTIVITIES	Actor 1. User opens the home page and searches by user or course.	System 1.1. System returns the profile of that user or information of that course.
EXCEPTION CONDITIONS	---	

3.5.16 Add Course/Group Preferences (Ecom 2023)

USE CASE NAME	Add Course/Group Preferences				
SCENARIO	Admin can assign specific professor, TA for specific course.				
TRIGGERING EVENT	When the faculty makes the schedule.				
BRIEF DESCRIPTION	The admin can assign a specific professor, TA for a specific course.				
ACTORS	Student Affairs				
RELATED USE CASES	Login User				
STAKEHOLDERS	Student Affairs				
PRECONDITIONS	Professor or TA for this course is active.				
POSTCONDITIONS	This professor takes this course.				
FLOW OF ACTIVITIES	<table> <tr> <th>Actor</th><th>System</th></tr> <tr> <td>1. The admin shows the schedule for professors to know who is active and assign the professors of course.</td><td>1.1. Every professor shows his courses.</td></tr> </table>	Actor	System	1. The admin shows the schedule for professors to know who is active and assign the professors of course.	1.1. Every professor shows his courses.
Actor	System				
1. The admin shows the schedule for professors to know who is active and assign the professors of course.	1.1. Every professor shows his courses.				
EXCEPTION CONDITIONS	---				

3.5.17 Login User (Ecom 2023)

USE CASE NAME	Login User								
SCENARIO	User enters his/her email and password so he/she can use the updated new-ecom.								
TRIGGERING EVENT	---								
BRIEF DESCRIPTION	User entered his email and password so he could use updated ecom.								
ACTORS	Student Affairs, Student, Professor, and TA								
RELATED USE CASES	---								
STAKEHOLDERS	Student Affairs, Student, Professor, and TA								
PRECONDITIONS	---								
POSTCONDITIONS	Users can use the allowed features.								
FLOW OF ACTIVITIES	<table> <tr> <th>Actor</th><th>System</th></tr> <tr> <td>1. The user uses the URL of the site.</td><td></td></tr> <tr> <td>2. The user enters the email and the password.</td><td></td></tr> <tr> <td></td><td>2.1. System displays the home page.</td></tr> </table>	Actor	System	1. The user uses the URL of the site.		2. The user enters the email and the password.			2.1. System displays the home page.
Actor	System								
1. The user uses the URL of the site.									
2. The user enters the email and the password.									
	2.1. System displays the home page.								
EXCEPTION CONDITIONS	2.1. User enters wrong password or email.								

3.5.18 Add Student (Ecom 2023)

USE CASE NAME	Add Student				
SCENARIO	Administrators can add new students to the system.				
TRIGGERING EVENT	when a new student needs to be added to the system, such as at the beginning of a new academic term.				
BRIEF DESCRIPTION	The administrator has the capability to add a new student to the system, providing necessary information for enrollment and record-keeping purposes.				
ACTORS	Administrator				
RELATED USE CASES	Login User				
STAKEHOLDERS	Student Affairs				
PRECONDITIONS	The administrator has appropriate permissions to add students to the system.				
POSTCONDITIONS	The new student is successfully added to the system, and their information is stored for future reference and management.				
FLOW OF ACTIVITIES	<table> <tr> <th>Actor</th><th>System</th></tr> <tr> <td>1. The administrator navigates to the "Add Student" functionality within the system.</td><td>1.1. The system presents a form or interface for entering student information.</td></tr> </table>	Actor	System	1. The administrator navigates to the "Add Student" functionality within the system.	1.1. The system presents a form or interface for entering student information.
Actor	System				
1. The administrator navigates to the "Add Student" functionality within the system.	1.1. The system presents a form or interface for entering student information.				
EXCEPTION CONDITIONS	If the entered student information is incomplete or inaccurate, the system notifies the administrator and prompts for corrections.				

3.5.19 Add Announcement (Ecom 2023)

USE CASE NAME	Add Announcement				
SCENARIO	Administrators can create and publish announcements to inform students, faculty, and staff about important events, updates, or news.				
TRIGGERING EVENT	when there is a need to announce information to the students				
BRIEF DESCRIPTION	The administrator has the capability to compose announcements and make them available to the students.				
ACTORS	Administrator				
RELATED USE CASES	Login User, View Announcement				
STAKEHOLDERS	Student Affairs				
PRECONDITIONS	The administrator has appropriate permissions to create and publish announcements.				
POSTCONDITIONS	The announcement is successfully created and published within the system, and users can access it for information.				
FLOW OF ACTIVITIES	<table> <tr> <th>Actor</th><th>System</th></tr> <tr> <td>1. The admin accesses the "Add Announcement" functionality within the system and writes the announcement and then submits it.</td><td>1.1. The system confirms successful publication of the announcement.</td></tr> </table>	Actor	System	1. The admin accesses the "Add Announcement" functionality within the system and writes the announcement and then submits it.	1.1. The system confirms successful publication of the announcement.
Actor	System				
1. The admin accesses the "Add Announcement" functionality within the system and writes the announcement and then submits it.	1.1. The system confirms successful publication of the announcement.				
EXCEPTION CONDITIONS	---				

3.5.20 Add Schedule File (Ecom 2023)

USE CASE NAME	Add Schedule File				
SCENARIO	Administrators can upload schedule files, such as academic calendars or course schedules.				
TRIGGERING EVENT	when there is a need to distribute schedule information to students, doctors, or TA				
BRIEF DESCRIPTION	The administrator has the capability to upload schedule files to the system.				
ACTORS	Administrator				
RELATED USE CASES	Login User, View Schedule				
STAKEHOLDERS	Student Affairs				
PRECONDITIONS	The administrator has appropriate permissions to upload schedule files.				
POSTCONDITIONS	The schedule file is successfully uploaded and stored within the system				
FLOW OF ACTIVITIES	<table> <tr> <th>Actor</th><th>System</th></tr> <tr> <td>1. The admin accesses the "Add Schedule File" functionality within the system and upload the schedule file.</td><td>1.1. The system confirms successful upload and provides users with access to the schedule file(s).</td></tr> </table>	Actor	System	1. The admin accesses the "Add Schedule File" functionality within the system and upload the schedule file.	1.1. The system confirms successful upload and provides users with access to the schedule file(s).
Actor	System				
1. The admin accesses the "Add Schedule File" functionality within the system and upload the schedule file.	1.1. The system confirms successful upload and provides users with access to the schedule file(s).				
EXCEPTION CONDITIONS	If the uploaded schedule file(s) are corrupted or incompatible with the system, the system notifies the administrator and prevents upload until the issue is resolved				

3.5.21 Accept New Students

USE CASE NAME	Accept New Students				
SCENARIO	Students apply for admission to the collage for the upcoming academic term				
TRIGGERING EVENT	When new students apply for admission				
BRIEF DESCRIPTION	Student affairs employees facilitates the process of admitting new students to the collage.				
ACTORS	Student Affairs staff				
RELATED USE CASES	-----				
STAKEHOLDERS	Student Affairs				
PRECONDITIONS	The admission system is operational, and Student Affairs staff have appropriate permissions to manage admission processes.				
POSTCONDITIONS	New students are successfully admitted to the college, and their admission information is recorded in the system.				
FLOW OF ACTIVITIES	<table> <thead> <tr> <th>Actor</th><th>System</th></tr> </thead> <tbody> <tr> <td>1. The student affairs employee gathers the details of the student's information in an Excel file and uploads it to the system.</td><td>1.1. The system confirms successful upload of the file and maps the necessary information to the model in the system.</td></tr> </tbody> </table>	Actor	System	1. The student affairs employee gathers the details of the student's information in an Excel file and uploads it to the system.	1.1. The system confirms successful upload of the file and maps the necessary information to the model in the system.
Actor	System				
1. The student affairs employee gathers the details of the student's information in an Excel file and uploads it to the system.	1.1. The system confirms successful upload of the file and maps the necessary information to the model in the system.				
EXCEPTION CONDITIONS	-----				

3.5.22 Issue Letter for various purposes.

USE CASE NAME	Issue Letter for Various Purposes	
SCENARIO	Student Affairs employees generate and issue letters to students for different purposes, such as Proof of registration, ..etc.	
TRIGGERING EVENT	when Students request official letters.	
BRIEF DESCRIPTION	Student Affairs employees generate and distribute official letters to students based on their requests or as required by institutional policies and procedures.	
ACTORS	Student Affairs Staff, Students	
RELATED USE CASES	-----	
STAKEHOLDERS	Student Affairs, Students	
PRECONDITIONS	Student Affairs staff have appropriate permissions to access and generate letters when the student pays for the service.	
POSTCONDITIONS	letters are successfully issued to students	
FLOW OF ACTIVITIES	Actor	System
	1. The student pays for the letter he is willing to get. 2. The student affairs employee start working on creating the letter	1.1. The system notifies the student that his letter or paper is in progress. 2.1. The system notifies the student that the letter is Done and is available.
EXCEPTION CONDITIONS	-----	

3.5.23 Issue Reports & statistics.

USE CASE NAME	Issue Reports & statistics	
SCENARIO	Student Affairs employees generate and provide reports and statistics to the Vice Dean for various purposes	
TRIGGERING EVENT	When the Vice Dean requests specific reports or statistics from Student Affairs employees	
BRIEF DESCRIPTION	Student Affairs employees present relevant data and information in the form of reports and statistics to Vice Dean.	
ACTORS	Student Affairs Staff, Vice Dean	
RELATED USE CASES	-----	
STAKEHOLDERS	Student Affairs, Vice Dean	
PRECONDITIONS	Student Affairs staff have appropriate permissions to access and generate reports.	
POSTCONDITIONS	Reports and statistics are successfully provided to the Vice Dean.	
FLOW OF ACTIVITIES	Actor - The student affairs employee receives the request from the vice dean for issuing or making a report - Once the student affairs finish the report, he sends it to the vice dean	System - The system presents a list of available reports templates for generating the requested reports or statistics. - The system notifies the vice dean that the reports is completed
EXCEPTION CONDITIONS	-----	

3.5.24 Handle Student Request

USE CASE NAME	Handle Student Request						
SCENARIO	Student Affairs employees or Vice Dean process and address various requests submitted by students						
TRIGGERING EVENT	When students submit requests for assistance or services to the Student Affairs department						
BRIEF DESCRIPTION	Student Affairs receive and respond to student requests to ensure their needs are met and appropriate actions are taken.						
ACTORS	Student Affairs Staff, Vice Dean, Students						
RELATED USE CASES	-----						
STAKEHOLDERS	Student Affairs, Vice Dean, Students						
PRECONDITIONS	Student Affairs staff have appropriate permissions to access and manage student requests.						
POSTCONDITIONS	Student requests are successfully addressed, and appropriate actions are taken to fulfill their needs.						
FLOW OF ACTIVITIES	<table> <tr> <th>Actor</th><th>System</th></tr> <tr> <td>1. A student submits a request for assistance or a service to the student affairs employees or Vice Dean</td><td>1.1. The system assigns that following student needs an assistance or service needed.</td></tr> <tr> <td>2. The student affairs employee handle the request for the student</td><td></td></tr> </table>	Actor	System	1. A student submits a request for assistance or a service to the student affairs employees or Vice Dean	1.1. The system assigns that following student needs an assistance or service needed.	2. The student affairs employee handle the request for the student	
Actor	System						
1. A student submits a request for assistance or a service to the student affairs employees or Vice Dean	1.1. The system assigns that following student needs an assistance or service needed.						
2. The student affairs employee handle the request for the student							
EXCEPTION CONDITIONS	-----						

3.6 ERD

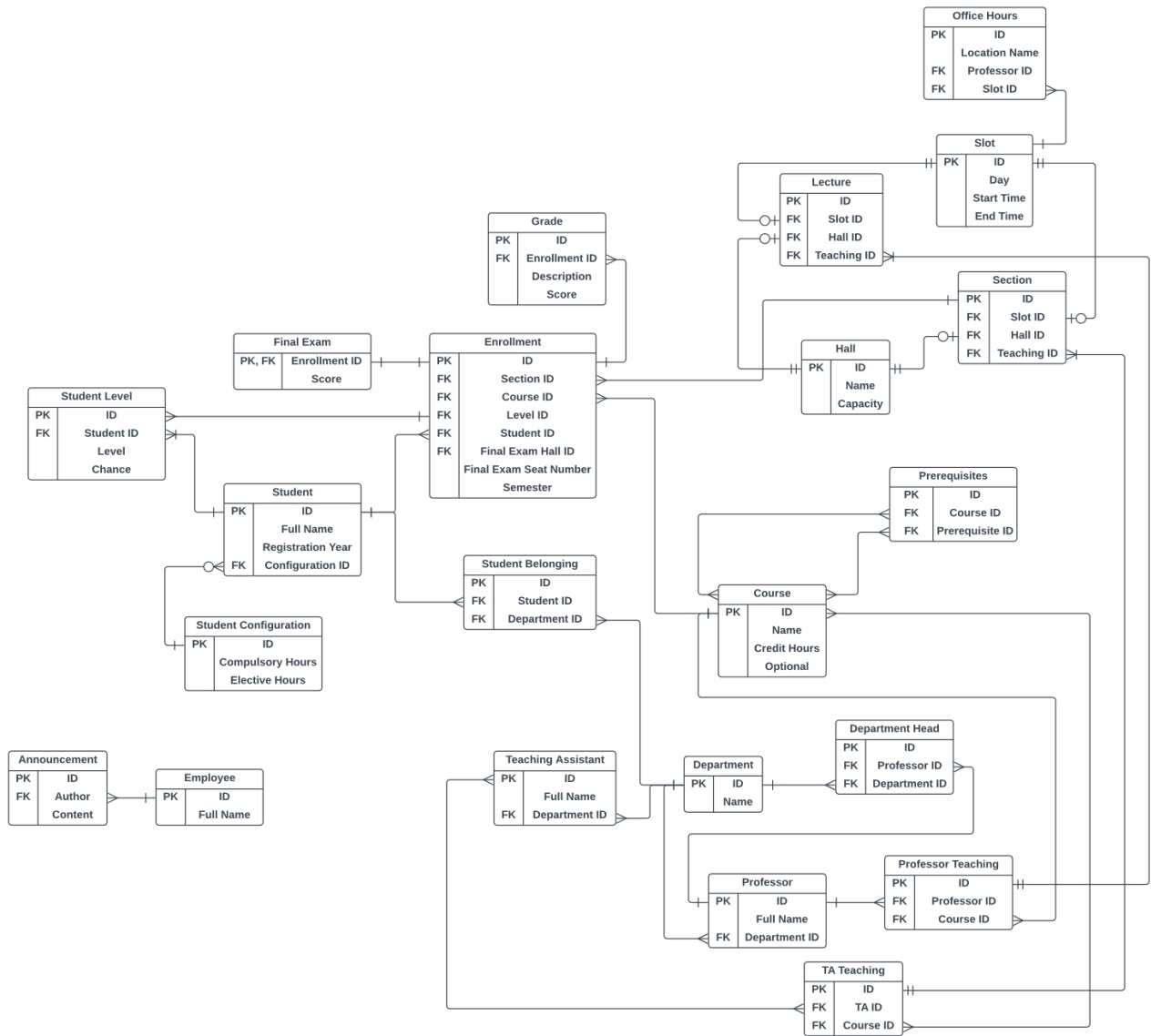


Figure 5. ERD

3.7 Component Diagram

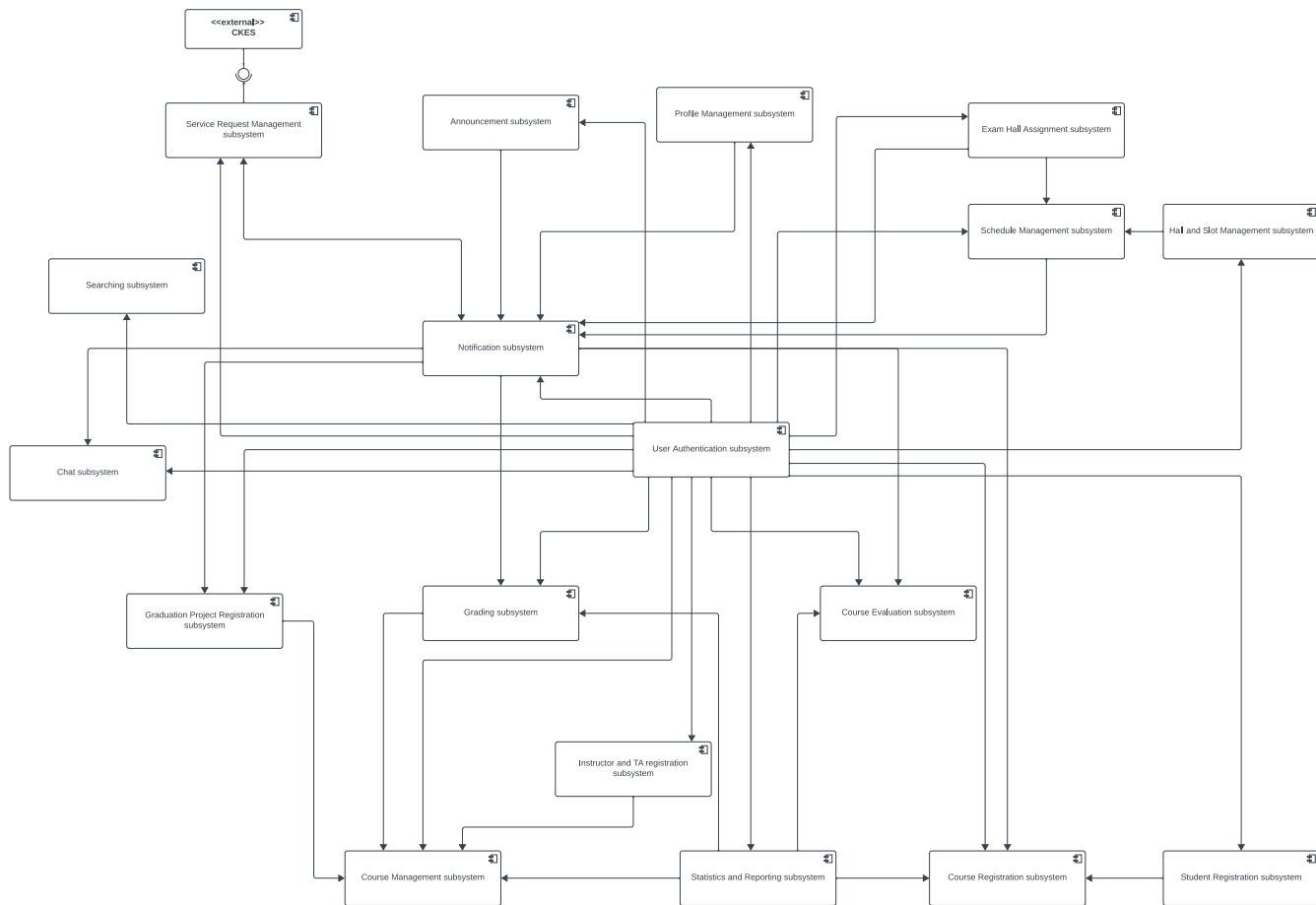


Figure 6. Component Diagram

3.8 Sequence Diagram

3.8.1 Register Course (Ecom 2023)

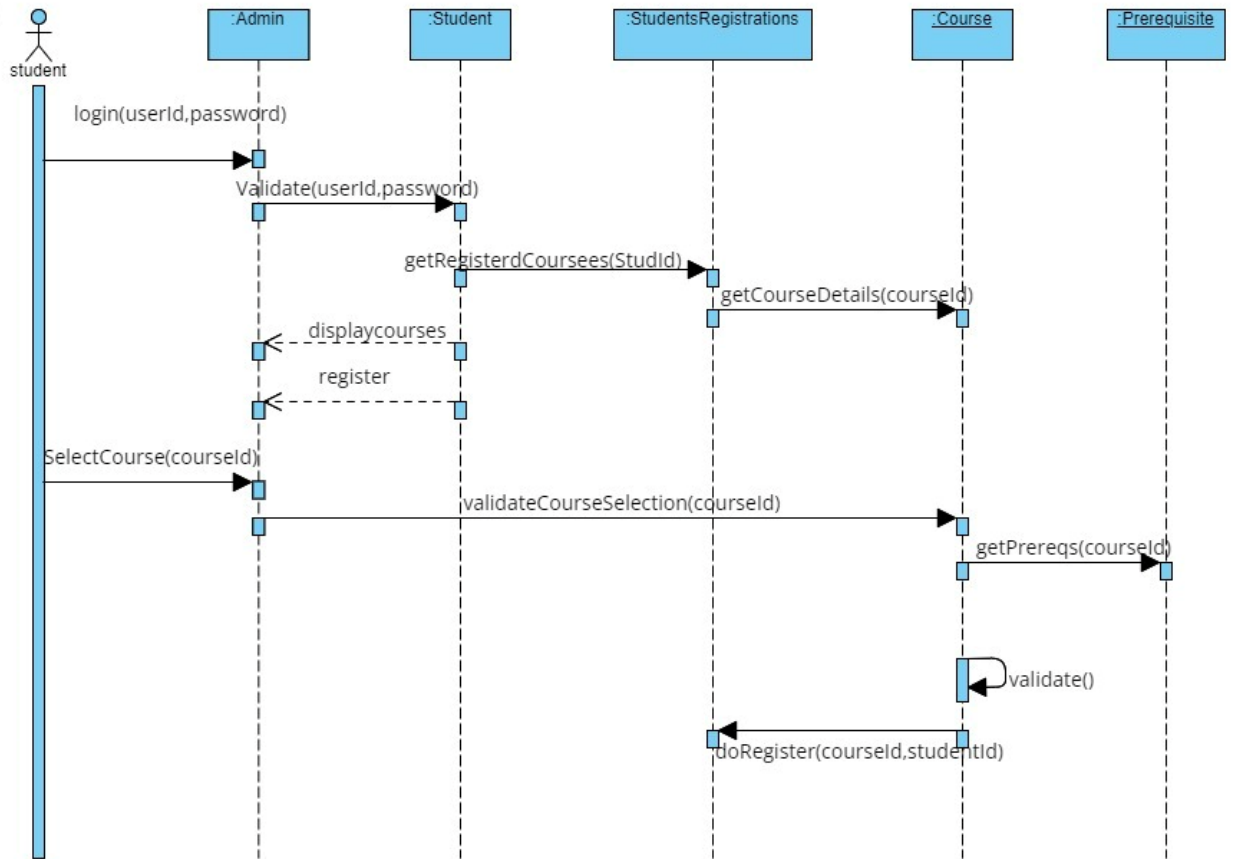


Figure 7. Register Course Sequence Diagram (Ecom 2023)

3.8.2 Internal Chat (Ecom 2023)

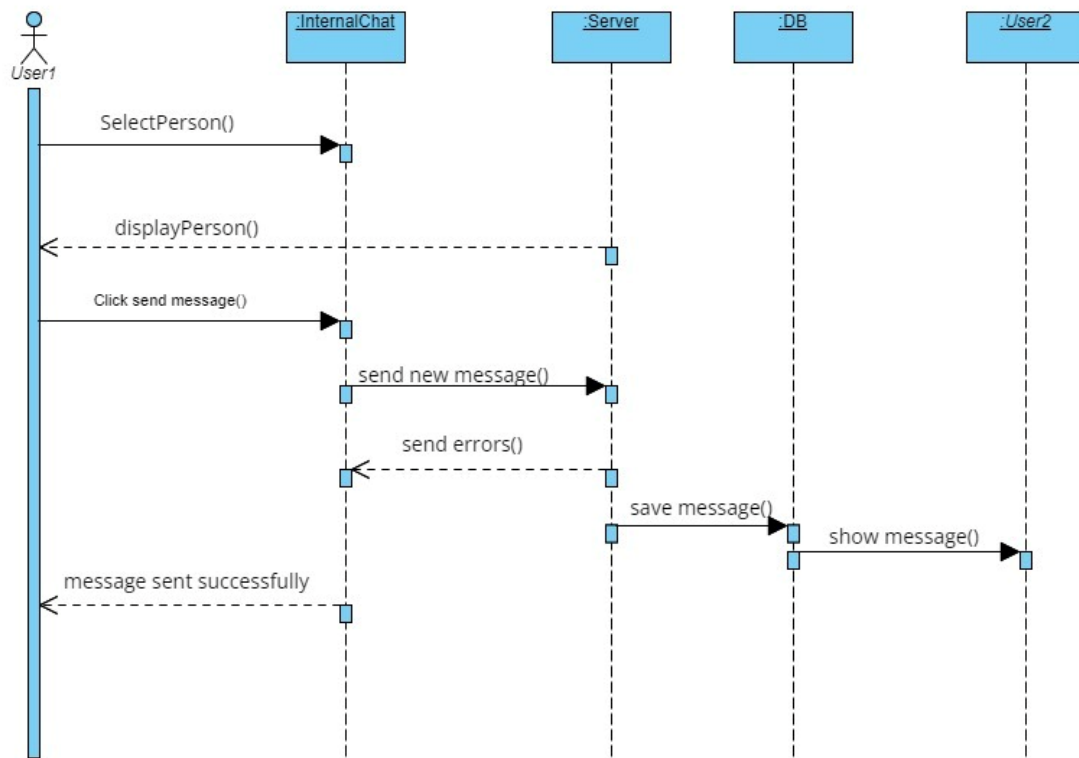


Figure 8. Internal Chat (Ecom 2023)

3.8.3 View Office Hours (Ecom 2023)

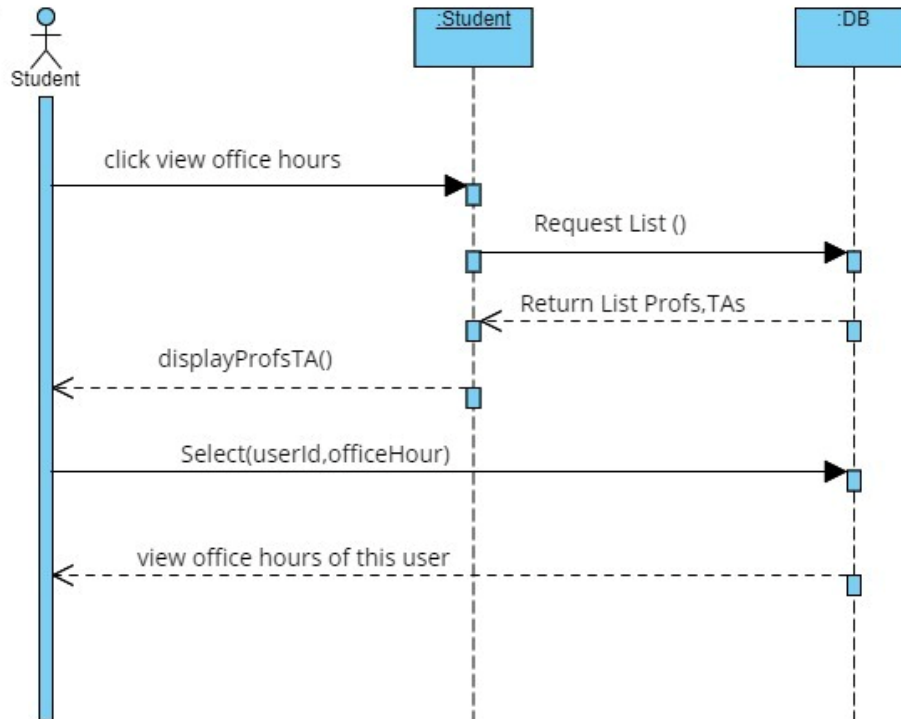


Figure 9. View Office Hours sequence Diagram (Ecom 2023)

3.8.4 View Schedule (Ecom 2023)

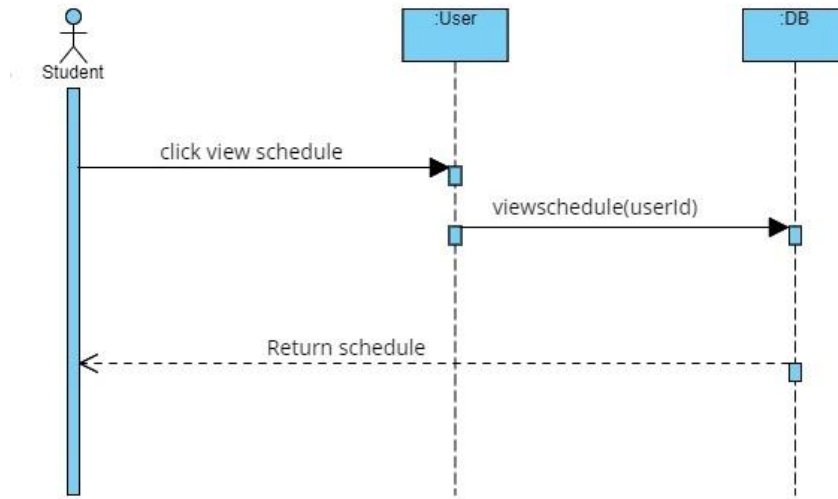


Figure 10. View Schedule Sequence Diagram (Ecom 2023)

3.8.5 Student Registration

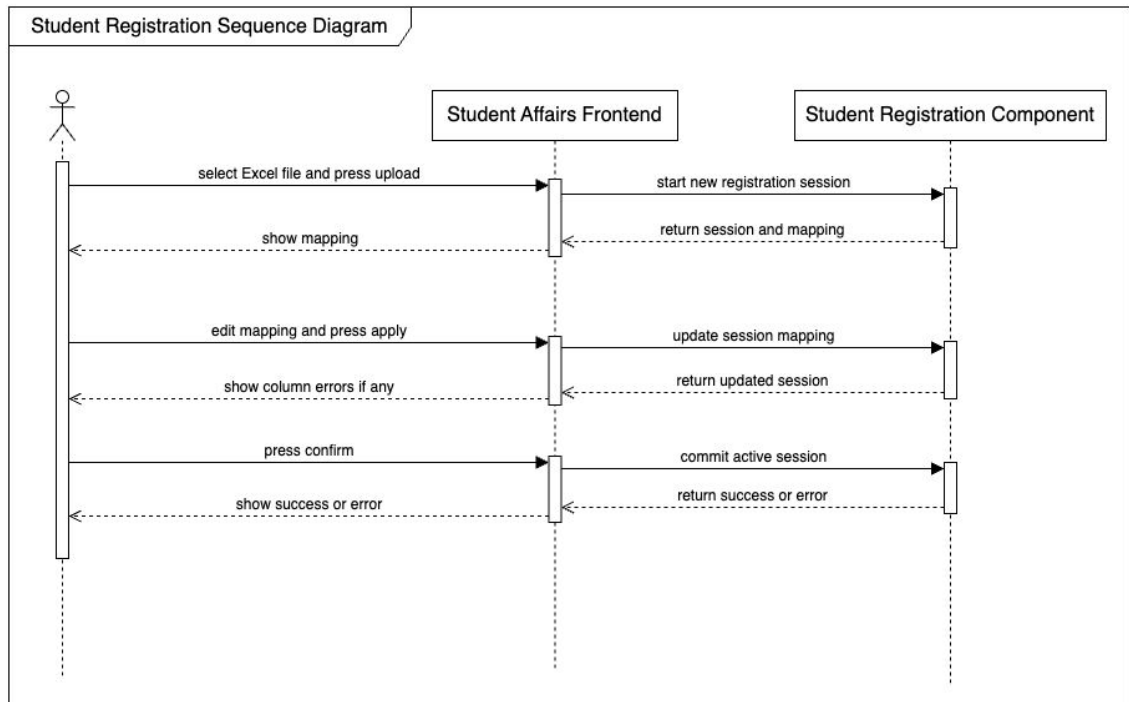


Figure 11. Student Registration Sequence Diagram

3.9 BPMN

3.9.1 Student Registration

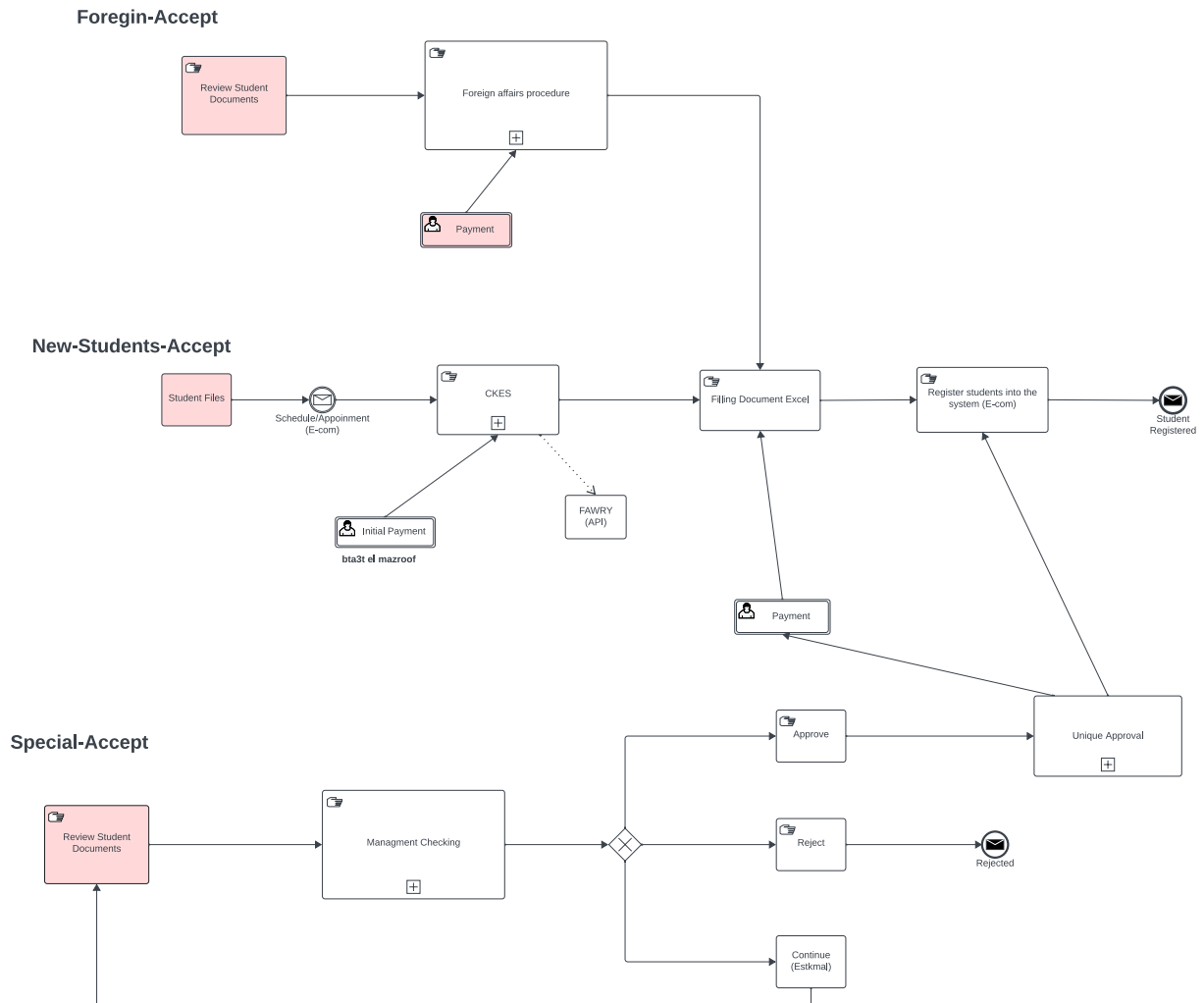
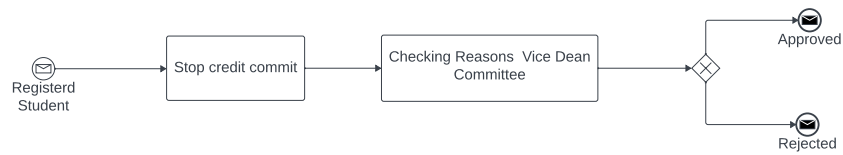
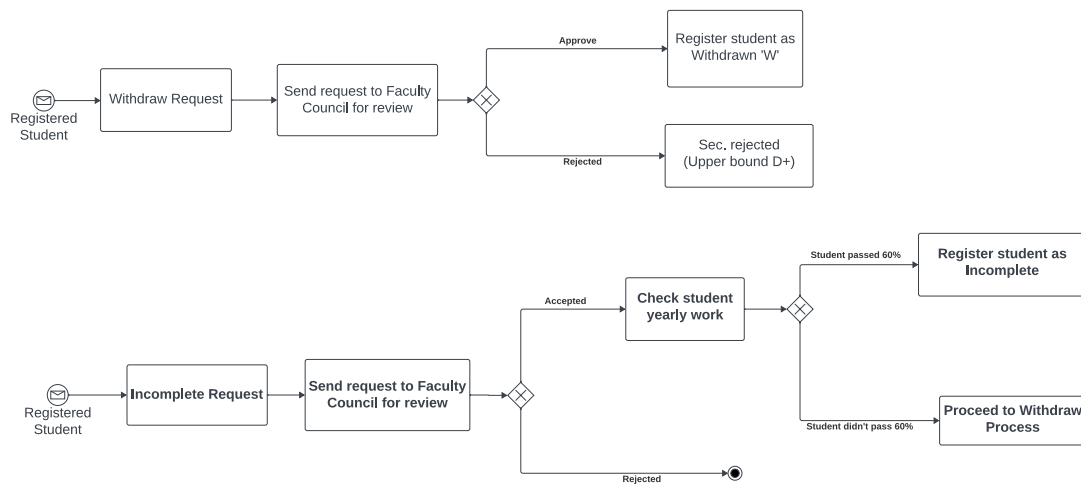


Figure 12. Student Accept

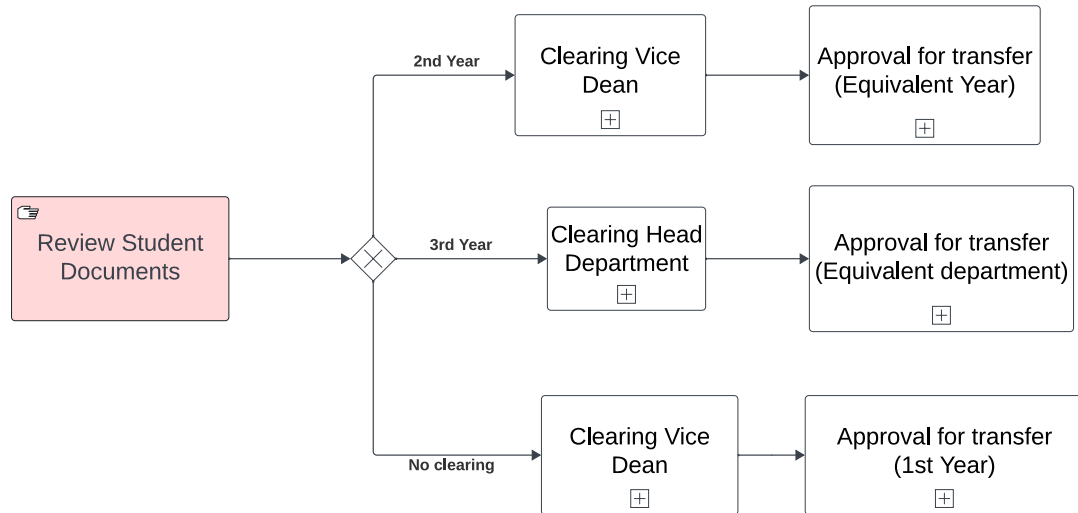
3.9.2 Stop Credential



3.9.3 Withdrawal/Incomplete



3.9.4 Clearing



Chapter 4 Design & Implementation

4.1 Technologies and Tools

Lucid Chart	Web-based diagramming application we used it for making component Diagram, use case diagrams, ERD, sequence diagrams and BPMNs.
Next.js Framework	React-based web applications with server-side rendering and static website generation.
Express Framework	Back-end web application framework for building RESTful APIs with Node.js.
GitHub	Developer platform for creating, storing, managing, and sharing our code.
Visual Studio Code	Source code editor.
Figma	Collaborative web application for interface design.
Hoppscotch.io	Web-based API development suite, for testing APIs.

4.2 System Screens

الاسم

اسم المستخدم

كلمة المرور

.....

تسجيل الدخول

بوابة شؤون الطلاب

لكلية الحاسبات والذكاء الاصطناعي
بجامعة القاهرة

English

Figure 13. Authentication screen

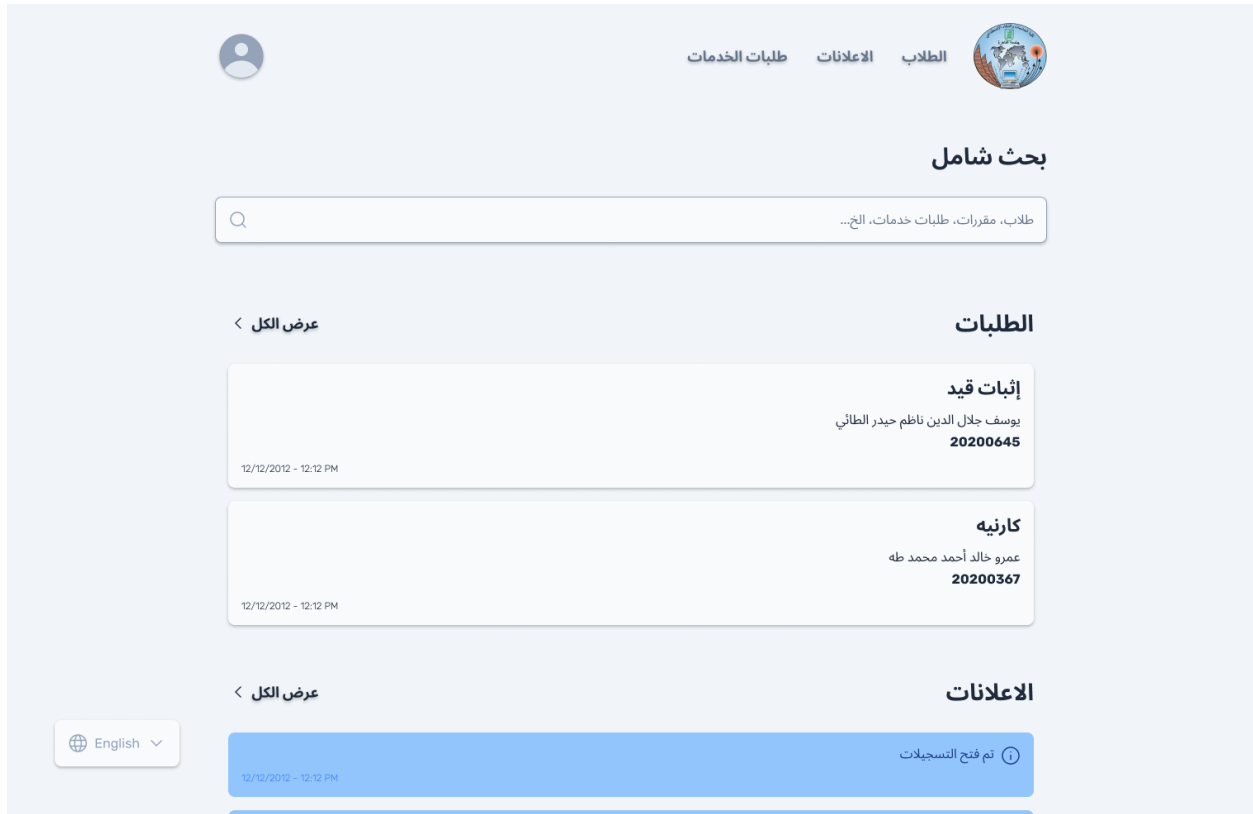


Figure 14. Home page

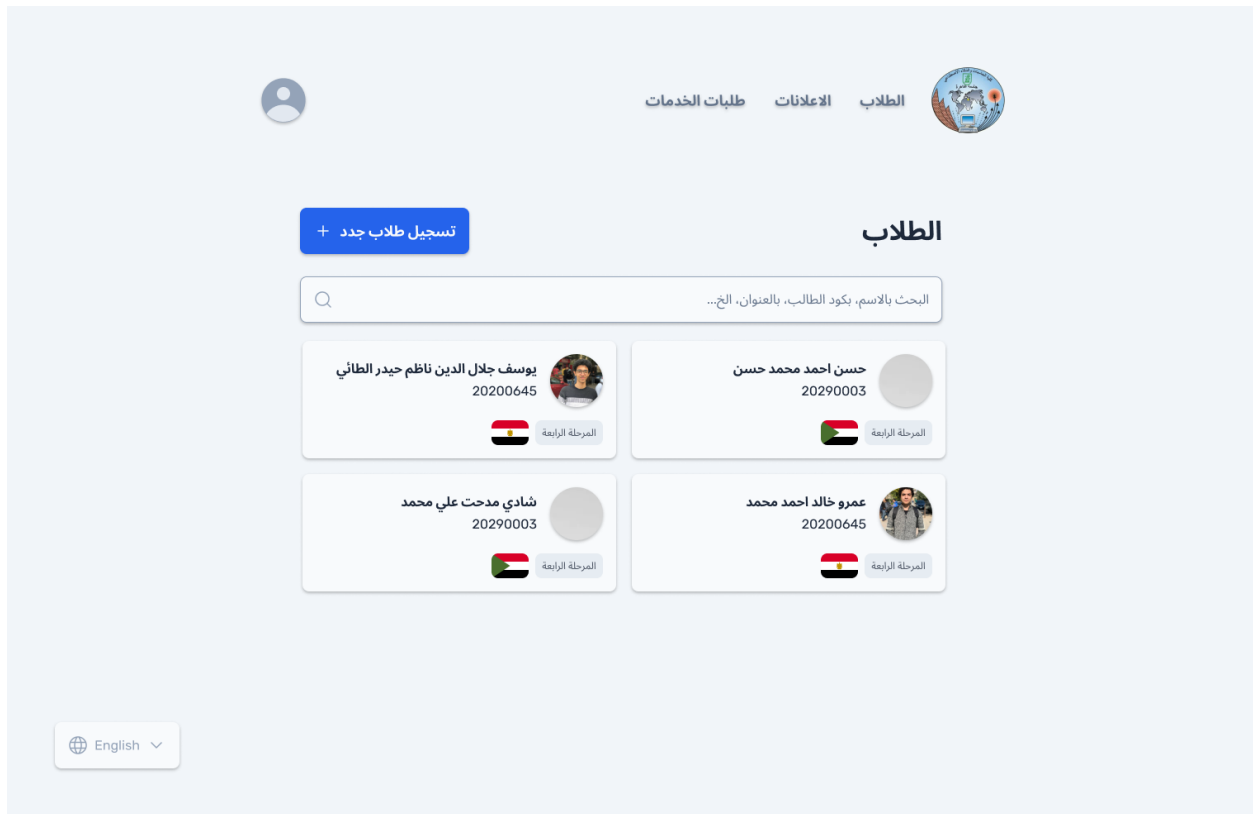


Figure 15. Students

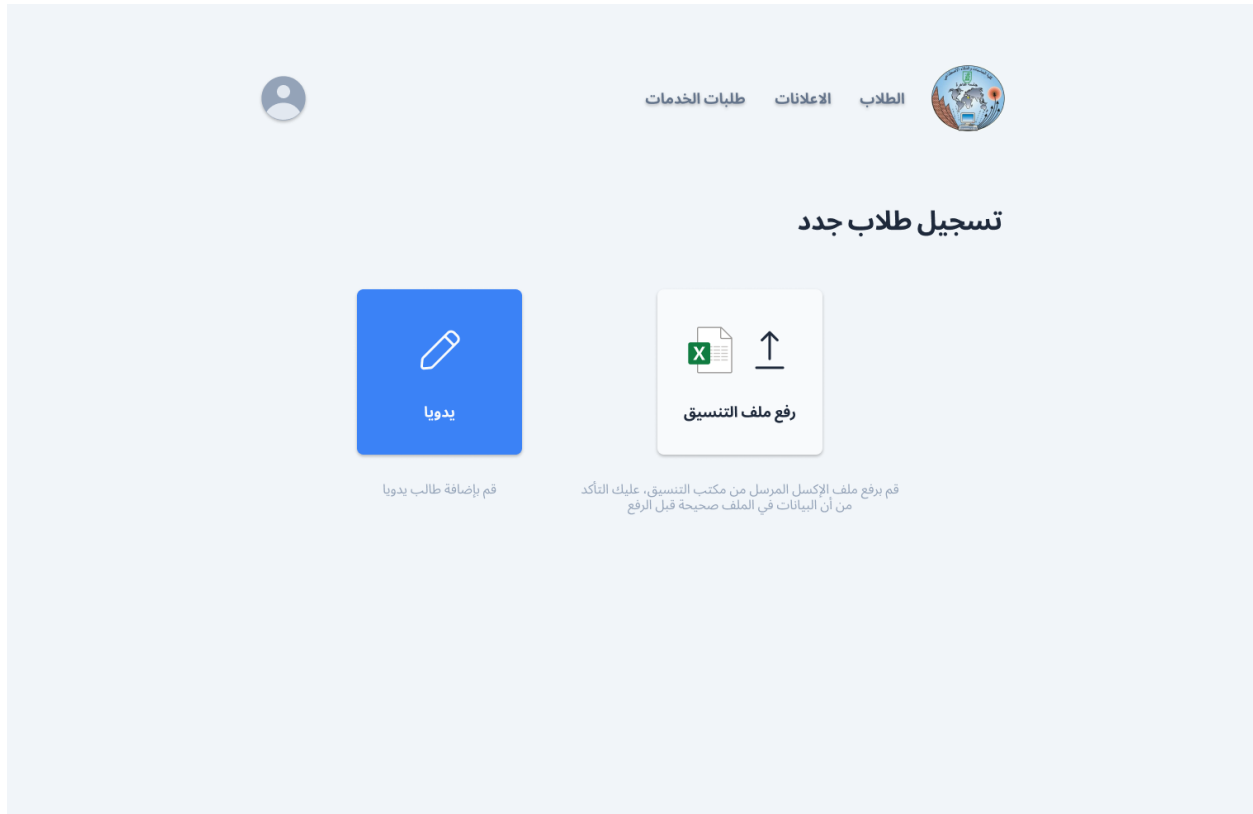


Figure 16. Register New Students.



الطلاب

الاعلانات

طلبات الخدمات



تسجيل طلاب من ملف التنسيق

٣

٢

١

مراجعة وتأكيـد

الربط بقاعدة البيانات

رفع الملف

من فضلك قم برفع ملف الإكسل المرسل من مكتب التنسيق بعد التأكد من أن البيانات في الملف صحيحة

اختر الملف

لم تقم باختيار ملف بعد

متابعة <

English

Figure 17. Choose Excel File

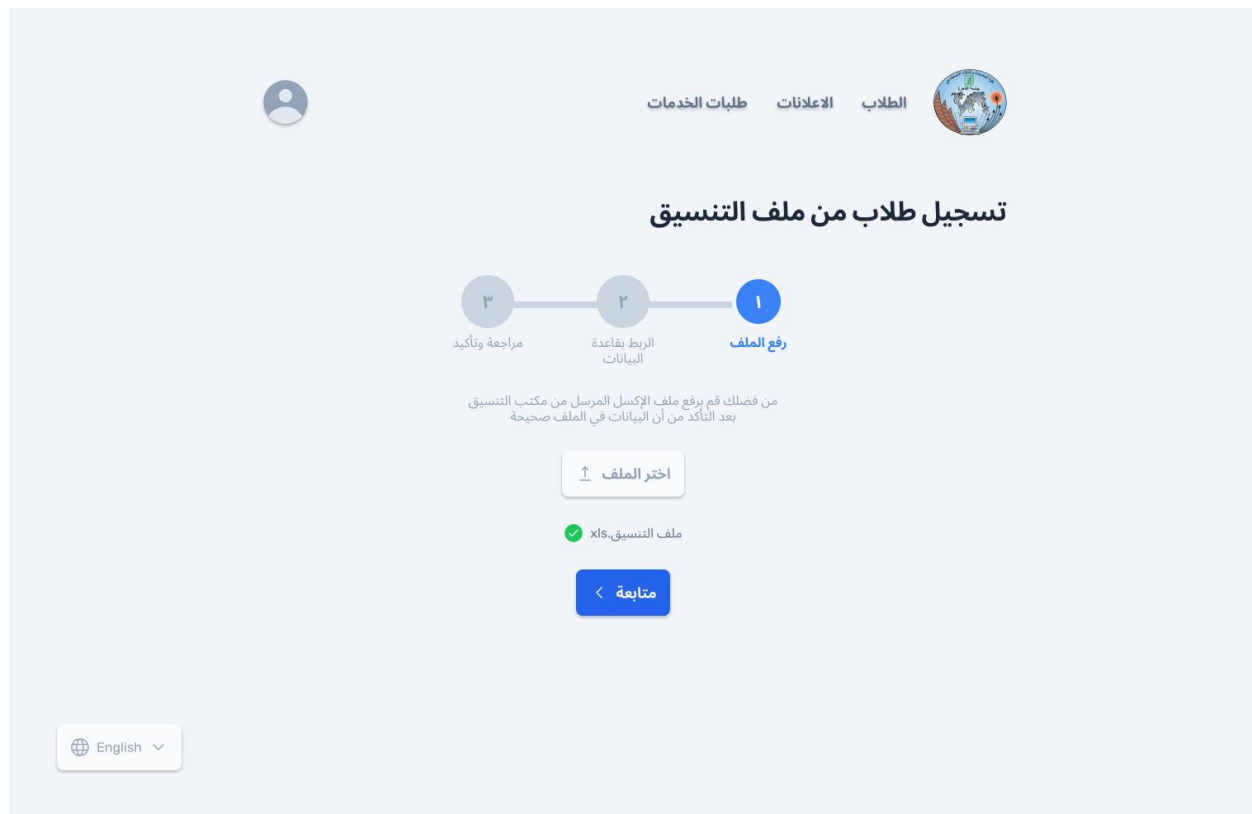


Figure 18. Upload Excel File

الطلاب

الاعلانات

طلبات الخدمات

٣

مراجعة وتأكيـد

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الربط بقاعدة البيانات

١

رفع الملف

من فضلك قم بربط الأعمدة الصحيحة في ملف الإكسل بالأعمدة المقابلة لها في قاعدة البيانات

اختـر عمودا	الرقم القومي
الاسـم	اسـم الطالب
جنسية	الجنسية

متابعة <

الرجوع >

English

Figure 19. Mapping-1

الطلاب

الاعلانات

طلبات الخدمات

٣

مراجعة وتأكيد

٢

الربط بقاعدة البيانات

١

رفع الملف

الرقم القومي

الاسم

البريد الإلكتروني

رقم الهاتف

الرقم القومي

الاسم

جنسية

الرقم القومي

اسم الطالب

الجنسية

من فضلك قم بربط الأعمدة الصحيحة في ملف الإكسل بالأعمدة المقابلة لها في قاعدة البيانات

متابعة <

الرجوع >

English

Figure 20. Mapping-2

58

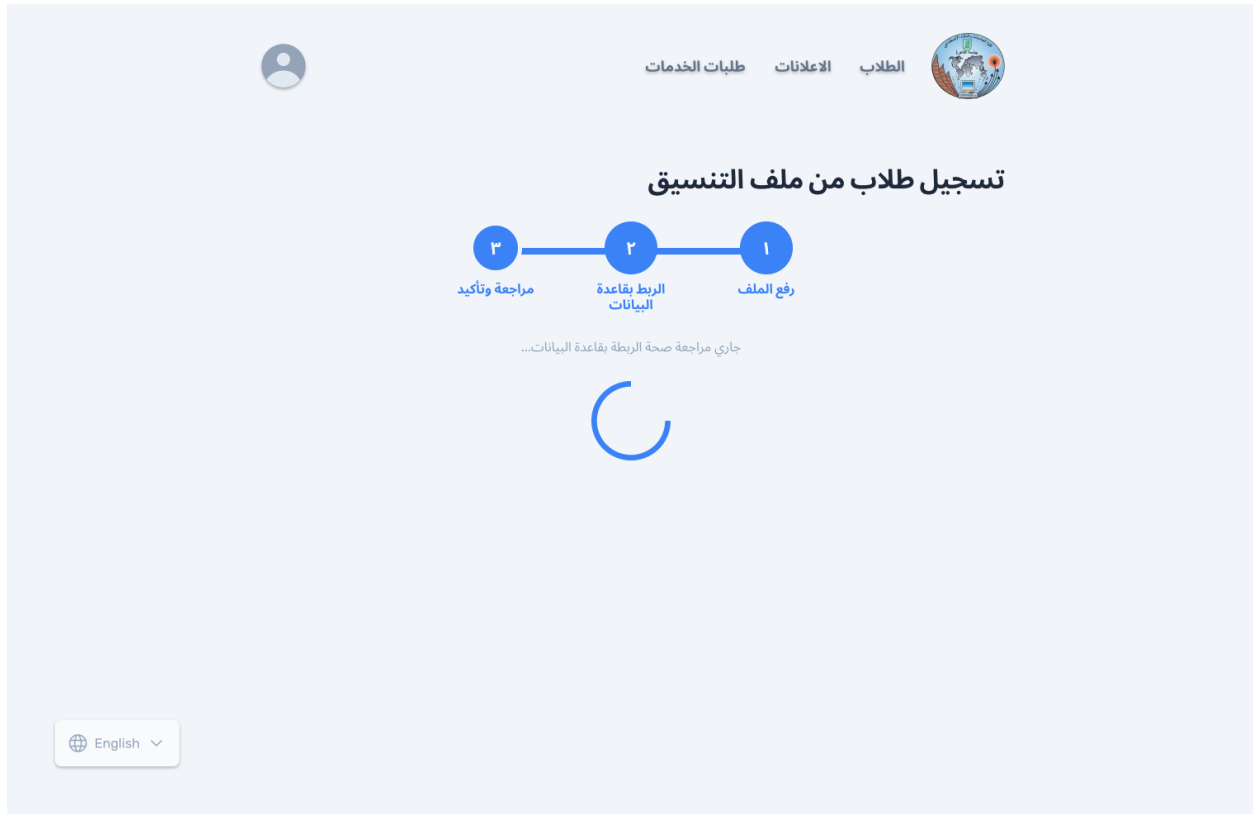


Figure 21. Loading Screen

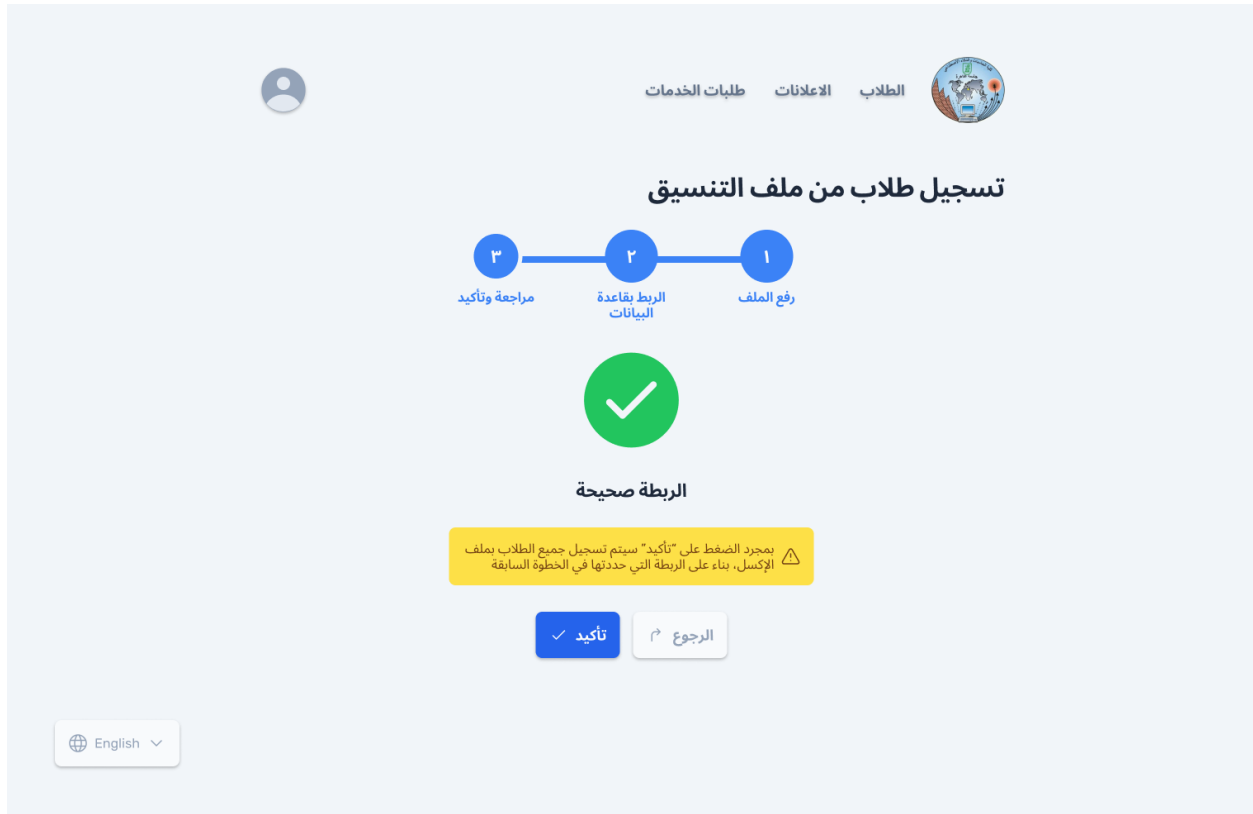


Figure 22. Confirmation